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SUPERSEDING
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FEDERAL SPECIFICATION

ADHESIVE, VINYL ACETATE RESIN EMULSION

This specificstion was approved by the Commissioner, Federal Supply Service General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of vinyl acetate resin emulsion adhesive for assembly gluing of wood items.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-G-460 - Glass Containers, One Gallon Capacity and Smaller for Other Than Medicinal Products; Packaging and Packing of.
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 through 12 Gallon).

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society For Testing And Materials (ASTM) Standards:

D905 - Strength Properties of Adhesive Bonds in Shear by Compression Loading.
D906 - Strength Properties of Adhesives in Plywood Type Construction in Shear by Tension Loading.
D1084 - Consistency of Adhesives.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. The adhesive shall be a vinyl acetate resin emulsion suitably modified to meet the requirements of this specification.

3.2 Viscosity. The viscosity of the adhesive shall be not less than 1000 nor more than 7000 centipoises when tested as specified in 4.5.1.

3.3 Shear strength (compression). The shear strength of the adhesive bond in hard maple block joints by compression loading shall be not less than 2800 p.s i. when tested as specified in 4.5.2.

3.4 Plywood shear strength tests.

3.4.1 Shear strength at 23° ± 1°C. (73.4° ± 1.8°F.). The shear strength of the adhesive bond in the plywood at 23° ± 1°C. (73.4° ± 1.8°F.), shall be not less than 400 p.s.i. when tested as specified in 4.5.3.1.1.

3.4.2 Shear strength at 71.1° ± 2.8°C. (160° ± 5°F.). The shear strength of the adhesive bond in the plywood at 71.1° ± 2.8°C. (160° ± 5°F.), shall be not less than 250 p.s.i. or 40 percent of the shear strength at 23° ± 1°C. (73.4° ± 1.8°F.) (see 4.5.3.1.1) whichever is larger when tested as specified in 4.5.3.1.2.

3.4.3 Shear strength after moisture exposure. The shear strength of the adhesive bond in the plywood after moisture exposure shall be not less than 250 p.s.i. or 40 percent of the dry strength (see 4.5.3.1.1), whichever is larger when tested as specified in 4.5.3.1.3.

3.5 Freeze-thaw-stability. After being subjected to 5 freeze-thaw cycles, the adhesive shall be capable of meeting all requirements of this specification. Tests shall be as specified in 4.5.4.

3.6 Storage life. After storage for 6 months from date of delivery, the adhesive shall be capable of meeting all requirements of this specification. Tests shall be as specified in 4.5.5.

3.7 Workmanship. The adhesive shall be homogeneous and free from lumps and foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of adhesive manufactured at one place from the same batch or blend of raw materials subjected to the same processing operations and conditions, and offered for delivery at one time.

4.3 Sampling.

4.3.1 For examination. Unless otherwise specified, a random sample of filled containers shall be selected for examination in accordance with level 1 of MIL-STD-105.

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4.3.2 For tests. A 2-quart sample of adhesive representative of the lot shall be selected for tests. One quart of the adhesive shall be placed in a wide-mouth, glass, 1-quart, container and sealed tightly for use for all tests except those of 4.5.4 and 4.5.5. Two 1-pint samples for tests shall be placed in containers as specified in 4.5.4 and if required, 4.5.5.

4.4 Examination. Sample units selected in accordance with 4.3.1 shall be examined for defects and at the acceptable quality level (AQL) shown in table I.

TABLE I. Classification of defects

Material	AQL percent	Classification of defect	Defect	Method of inspection
Adhesive (see 3.7)	2.5	Major 101	Not homogeneous	Visual
		Major 102	Not free from lumps	Visual
		Major 103	Contains foreign matter	Visual
Unit containers open (see 5.1)	2.5	Major 104	1/Improper fill	Visual
		Major 105	Improper type	Visual
		Major 106	Improper size	Visual
		Major 107	Improper coating for metal containers	Visual
Unit containers closed (see 5.1 and 5.3)	2.5	Major 108	Leakage	Visual
		Major 109	Improper closure	Visual
		Major 110	Improper marking	Visual
Interior boxes (squeeze bottles only) (see 5.1.1)	2.5	Major 111	Wrong size or type of paperboard box	Visual
		Major 112	Wrong number of squeeze bottles in box	Visual
		Major 113	Wrong type partition	Visual
Box open (see 5.2)	2.5	Major 114	Improper size	Visual
(For packing of squeeze 2/ bottles and metal cans only)		Major 115	Improper type	Visual
		Major 116	Lack of, or improper strapping	Visual
Box closed (see 5.2 and 5.3)	2.5	Major 117	Gross weight max.	3/Approved scale
(For packing of squeeze 2/ bottles and metal cans only)		Major 118	Improper marking	Visual
		Major 119	Improper closed	Visual

1/The fill of the containers shall be determined by comparison of the weight with the weight of a container filled with the minimum required quantity of adhesive.

2/Packing of glass containers. Examination of packing of glass containers shall be in accordance with the defects and AQL as specified in PPP-G-460.

3/Approved by procuring agency.

4.5 Tests.

4.5.1 Viscosity. The viscosity of the adhesive shall be determined as specified in ASTM D 1084 method B.

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4.5.2 Shear strength (compression). The shear strength of the adhesive bond by compression loading shall be determined in accordance with ASTM D 905.

4.5.3 Plywood shear tests.

4.5.3.1 Preparation of specimens. Specimens for the plywood shear tests of 4.5.3.1.1, 4.5.3.1.2 and 4.5.3.1.3 shall be prepared and tested in accordance with ASTM D 906 with the following exceptions: Sixty specimens (6 panels) shall be prepared. The 60 specimens shall be randomly separated into 4 groups of 15 specimens each. One group shall be tested in accordance with 4.5.3.1.1, the second group in accordance with 4.5.3.1.2, and the third group in accordance with 4.5.3.1.3. The fourth group shall be held for a reference test, in case one of the other groups fail test.

4.5.3.1.1 Shear strength at 23° + 1°C. (73.4° + 1.8°F.). Specimens prepared as specified in 4.5.3.1 shall be conditioned at 23° + 1°C. (73.4° + 1.8°F.), and relative humidity of 50 + 2 percent as specified in ASTM D 906 and tested as specified in D 906.

4.5.3.1.2 Shear strength at 71.1° + 2.8°C. (160° + 5°F.). The jaws of the testing apparatus used in 4.5.3.1.1 shall be enclosed in a chamber capable of maintaining an air temperature of 71.1° + 2.8°C. (160° + 5°F.). Each specimen shall be kept unloaded in the chamber at this temperature for 30 + 2 mins. and then rapidly place in the jaws of the machine and tested at the same temperature. The procedure for testing shall be as described in ASTM D 906.

4.5.3.1.3 Shear strength after moisture exposure. Specimens prepared as specified in 4.5.3.1 shall be exposed for one week to an atmosphere maintained at 23° + 1°C. (73.4° + 1.8°F.) and 90 + 2 percent relative humidity. In this exposure the specimens shall be placed uniformly to permit good air circulation between them. Following the exposure period, the specimens shall be removed individually from the exposure chamber and tested immediately at normal room conditions as specified in ASTM D 906, with precautions being taken to avoid any appreciable change in moisture content during test.

4.5.4 Freeze-thaw stability. One pint of the adhesive shall be placed in a suitable container and stored at a temperature of -28.9° + 2°C. (-20° + 3.6°F), for 18 hours. The sample shall be restored to a temperature of 23° + 1°C. (73.4° + 1.8°F.), without stirring. The sample shall then be subjected to 4 additional freeze-thaw cycles. Following these freeze-thaw cycles, the sample shall be tested for the requirements of 3.2 and 3.4.1, 3.4.2 and 3.4.3.

4.5.5 Storage life. Unless otherwise specified (see 4.5.5.1), a representative can of adhesive of the lot offered for inspection shall be stored at a temperature of 23° + 1.1°C. (73.4° + 2°F.), for 6 months. At the end of this period the adhesive shall be tested for the requirements of 3.2, 3.4.1, 3.4.3, and 3.4.3.

4.5.5.1 Certification. When specified (see 4.5.5), the supplier shall submit to the procuring agency a letter certifying that the adhesive has met (see 4.5.5.1.1) or shall meet. as applicable, the storage life requirements of 3.6. The letter shall be signed by a responsible agent of the certifying organization and shall be accompanied by evidence of this agent's authority to bind his principal.

4.5.5.1.1 Storage for certification. When storage of the adhesive is performed by the supplier for certification, storage at a temperature of 16° to 32°C. (60° to 90°F.), will be considered equivalent to the specified temperature. In case of disagreement between the certifying organization and the Government, however, storage shall be in accordance with 4.5.5

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Adhesive furnished in 2-ounce, 1-pint, 1-quart, 1-gallon, and 5-gallon quantities as specified (see 6.2), shall be packaged as follows: Two-ounce quantities shall be packaged in commerical, plastic squeeze-type bottles. Twelve plastic squeeze bottles shall be packaged in a paperboard box conforming to PPP-B-566, style II, type A, class A, containing half-slotted, full height partitions forming an individual cell for each bottle. The 1-quart containers shall conform to type V, class 2 of PPP-C-96 or glass jars. The 1-pint or 1-gallon containers shall conform to type V, class 2 of PPP-C-96. Unless otherwise specified, the metal containers shall have exterior protective coatings conforming to plan A. Five gallon containers shall conform to type II, class 3 of PPP-P-704 with olive drab exterior coatings. Closure of the 1-pint, 1-quart. and 1-gallon cans shall be in accordance with the appendix to PPP-C-96.

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5.1.2 Level C. The adhesive shall be packaged to afford adequate protection against deterioration and damage during shipment from the supplier to the initial destination.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Twelve boxes of plastic squeeze bottles (144 bottles) shall be packed in a fiberboard box conforming to PPP-B-636, class weather-resistant. Adhesive packaged in 1-quart glass jars shall be packed in accordance with level A requirements of PPP-G-460. Adhesive packaged in 1-pint, 1-quart, or 1-gallon cans shall be packed in accordance with the level A requirements of the appendix to PPP-C-96. Five gallon containers will require no over-packing.

5.2.2 Level B. Twelve boxes of plastic squeeze bottles (144 bottles) shall be packed in a fiberboard box conforming to PPP-B-636, class domestic. Adhesive packaged in 1-quart glass jars shall be packed in accordance with the level B requirements of PPP-G-460. Adhesive packaged in 1-pint, 1-quart, or 1-gallon cans shall be packed in accordance with level B requirements of the appendix to PPP-C-96. Five-gallon containers will require no over-packing.

5.2.3 Level C. The adhesive shall be packed, to insure arrival at destination in satisfactory condition, in a container complying with the rules and regulations of the applicable mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. Packages and shipping containers shall be marked in accordance with Fed. Std. No. 123 and the special marking of 5.3.2.

5.3.2 Special marking. In addition to the marking of 5.3.1 or 5.3.3 as applicable, and any marking required by the contract or order (see 6.2) information shall appear on each container as follows:

(a) Manufacturers instructions for use (to include thinning directions if applicable).
(b) Date of manufacture (by month and year, not by code).
(c) Any special directions for storage or use of the adhesive.
(d) In addition to the above the following instructions shall appear on each container:

"STORE IN A COOL PLACE.
KEEP FROM FREEZING."

5.3.3 Military agencies. Packages and shipping containers shall be marked in accordance with MIL-STD-129 and the special marking of 5.3.2

6. NOTES

6.1 Intended use. This adhesive is considered suitable for use in assembly gluing of wood items which will be placed or stored in normal indoor temperature service conditions and where the relative humidity is not high and does not fluctuate between wide limits. In application, this adhesive is intended for joints such as dowel, mortise-tenon, lock, and finger in the assembly of parts of wood furniture. It is also considered suitable for use in fabricating small wood patterns and models. This adhesive is not considered suitable for edge-gluing and laminating for furniture parts

6.2 Ordering data. Purchasers should select the prefered option permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Selection of the applicable levels of packaging and packing (see 5.1, 5.2, and 6.2.1).
(c) Size and type container required (see 5.1).
(d) Whether other than plan A exterior protective coating is required for level A, 1-pint, 1-quart, or 1-gallon containers (see 5.1.1).
(e) Special marking if required (see 5.3.2).

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6.2.1 For civil agency procurement. When level B packaging is required for civil agency procurements the requirements in 5.1.1, level A packaging shall apply.

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